🔧 Scytale Junior Data Engineer Home Assignment

Hi there!

In this home assignment, you'll be developing a script that integrates with GitHub to help ensure that pull requests (PRs) are merged only after passing review.

We're going to write a script to **fetch merged pull requests from a GitHub repository** and check if they passed review.

Make sure you research the relevant API documentation and understand the entities relationships.

📌 Requirements

Write a script that:

- Authenticates with GitHub using a **personal access token (PAT)**.

- Fetches **merged pull requests** from a **single GitHub repository** under the organization:

  👉 https://github.com/home-assistant

- Extract and process the data from the repository entities.
- Generate a report in CSV or Parquet format that will present:
  - For each merged pull request, check whether:

- The PR was **approved** by at least one reviewer?
- All **required status checks** passed before the merge?

You should organize your code into a standard application structure, containing the following:

- ai_utils/   (folder containing all AI tools resources (markdown files))
- extract.py: Responsible for retrieving the raw PR data from GitHub.
- transform.py: Responsible for manipulating and analyzing the data (e.g., computing whether checks and reviews passed).
- main.py
- utils.py #optional - if needed

Your script should generate outputs (e.g., extracts JSON) for each step into the appropriate subfolders.

📋 Your Report Should Include (Per PR)

- PR number.

- PR title.
- Author.
- Merge date.
- CR_Passed (Whether it passed code review).
- CHECKS_PASSED (Whether all required checks passed)

💎 Bonus - Agentic AI approach (Optional)

If you'd like to go further, here are some stretch goals:

- Write an AI prompt command to reproduce the entire project requirements outputs for another integration i.e. JIRA extracting the list of tickets using API key.

  Remember! having more than 1 integration supported will require shared functionalities/modules among integrations. (hint: OOP)

📘 What We're Looking For

- Working code that demonstrates thoughtfulness and correctness.
- Clear structure and code readability.
- Proper use of GitHub's API and handling of edge cases.
- Bonus: Modular and reusable code design.

📨 Submission Instructions

Please submit a GitHub repository with your project in an organized structure.

If anything is unclear or you're stuck, reach out to us via email or phone- we're happy to help.

Good luck and enjoy the challenge!